Language Models for Cold-Start Recommenders

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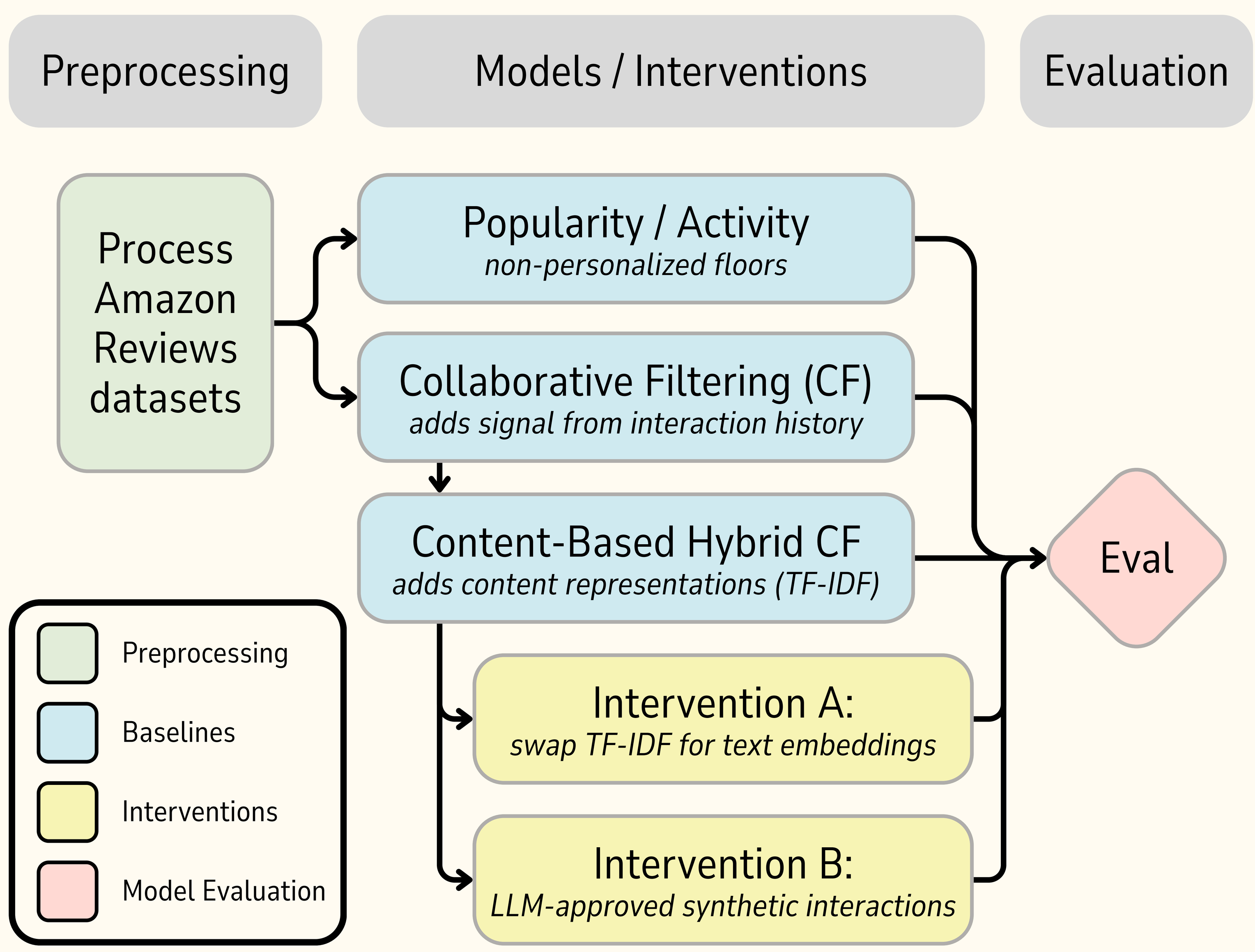
Introduction

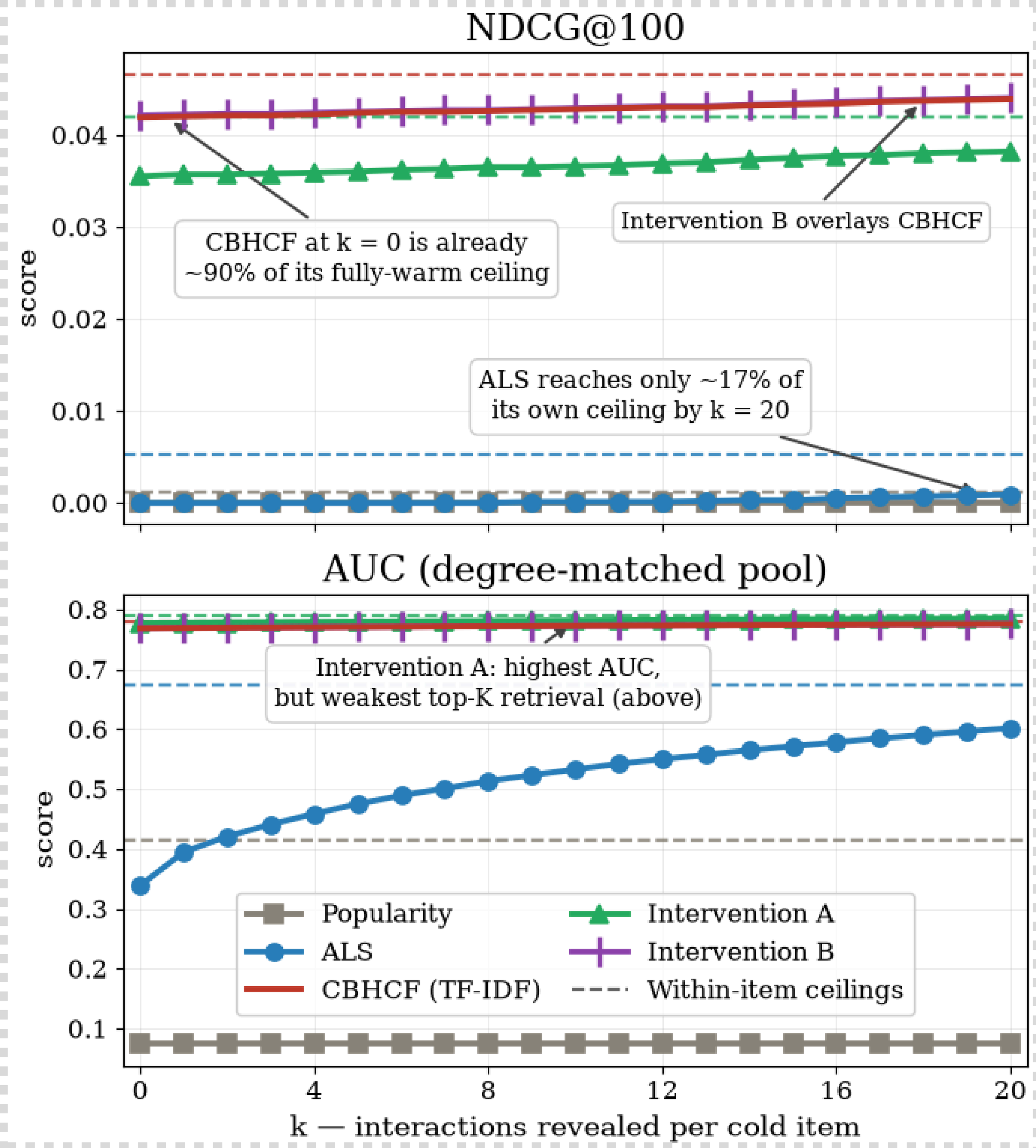
Cold items are a challenge for recommender systems. Pure collaborative filtering depends on interaction history and cannot rank fully cold items at all. Using a subset of the Amazon Reviews Books dataset (~2.67M interactions, 246,687 items), we evaluate recovery rather than a single snapshot, adding interactions one at a time to trace a **warm-up curve** showing how quickly an item becomes findable. We measure it from two directions: **Mode A** ranks items for a user, the familiar recommendation task, and **Mode B** recommends users for an item, informing publisher tasks such as targeted marketing. We test two LLM-assisted interventions against a TF-IDF content hybrid (CBHCF), our true baseline, and track **provider-side equity**, which captures whether smaller providers and newer items get a fair shot at exposure. We compare each model to a naive floor and a fully-warm ceiling.

Methodology



Purpose & Learning Objectives

Our goal is to quantify how much interaction history an item needs before a recommender performs well, and whether two interventions can reduce it. Intervention A replaces the hybrid's TF-IDF content representation with dense semantic embeddings. Intervention B adds LLM-generated synthetic interactions, controlled against random selection from the same candidate shortlist. Each intervention changes one thing and holds the rest fixed; user preferences are fit once, frozen, and reused identically for every method, so a change in retrieval quality is attributable to the item gaining signal.



Findings

Both language-model interventions added little beyond what item content had already provided, though for different reasons. Intervention B produced the highest NDCG in our comparison (0.04206 vs 0.04192 for CBHCF, a 0.33% lift), but most of that wasn't the LLM: random selection from the same content-filtered shortlist captured about two-thirds of the lift, leaving 0.11% attributable to the LLM's judgment. This was statistically significant, but negligible in practice, and selecting the most active users beat both at ~0.4%. Intervention A exposed a tradeoff between ranking and retrieval, with the best AUC but lower retrieval metrics on every top-K measure.

On provider equity, the content-based models spread exposure most evenly (Gini 0.8641–0.8784 vs 0.9950–0.9994 for the Popularity and CF baselines), although the flatter exposure didn't always lead to more cold-item exposure: Intervention A was the flattest but gave cold items a smaller share of exposure than CBHCF (0.0206 vs 0.0250).

Conclusion

Neither intervention, as we implemented it, improved on the TF-IDF content-based hybrid baseline. Intervention A's embeddings underperformed TF-IDF on retrieval while improving global ranking, but our claim is deliberately limited to the setup we tested. Our synthetic interactions came from an off-the-shelf 7B model without fine-tuning or the two-tower filtering of ColdLLM, so the result may be bounded by methodology rather than a ceiling on the approach. However, synthetic interaction generation should beat simple activity-based selection to earn its keep, and ours does not. Fine-tuning, improved filtering, and higher volume remain untested.

Select Reference:
more in report (see GitHub)

Huang, F., Bei, Y., Yang, Z., Jiang, J., Chen, H., Shen, Q., Wang, S., Karray, F., & Yu, P. S. (2024). Large Language Model Simulator for Cold-Start Recommendation. arXiv preprint arXiv:2402.09176. https://arxiv.org/abs/2402.09176

https://github.com/jessylan/llm-cold-start-recsys